more people, he's the same size as Mo and Pat, who are connected as many times but to fewer people.

CLUSTERING AND MODULARITY

One of the most important things to find out about a network is whether any communities exist in that network and what they look like. This is done by looking at whether some nodes are more connected to each other than to the rest of the network, known as *modularity*. You can also look at whether nodes are interconnected, known as *clustering*. Cliques, mentioned earlier, are part of the same measurement, and *clique* is a term for a group of nodes that are fully connected to each other.

Notice that this interconnectedness and community structure is supposed to arise visually out of a force-directed layout. You see the four highly connected users in a cluster and the other users farther away. If you'd prefer to measure your networks to try to reveal these structures, you can see an implementation of a community detection algorithm implemented by David Mimno with D3 at http://mimno.infosci.cornell.edu/community/. This algorithm runs in the browser and can be integrated with your network quite easily to color your network based on community membership.

6.2.4 Force layout settings

When we initialized our force layout, we started out with a charge setting of -1000. Charge and a few other settings give you more control over the way the force layout runs.

CHARGE

Charge sets the rate at which nodes push each other away. If you don't set charge, then it has a default setting of -30. The reason we set charge to -1000 was because the default settings for charge with our network would have resulted in a tiny network onscreen (see figure 6.12).

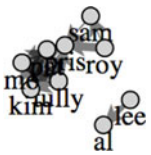


Figure 6.12 The layout of our network with the default charge, which displays the nodes too closely together to be easily read

Along with setting fixed values for charge, you can use an accessor function to base the charge values on an attribute of the node. For instance, you could base the charge on the weight (the degree centrality) of the node so that nodes with many connections push nodes away more, giving them more space on the chart.

Negative charge values represent repulsion in a force-directed layout, but you could set them to positive if you wanted your nodes to exert an attractive force. This would likely cause problems with a traditional network visualization but may come in handy for a more complicated visualization.

GRAVITY

With nodes pushing each other, the only thing to stop them from flying off the edge of your chart is what's known as *canvas gravity*, which pulls all nodes toward the center of the layout. When gravity isn't specifically set, it defaults to .1. Figure 6.13 shows the results of increasing or decreasing the gravity (from our original `charge(-1000)` setting).

Gravity, unlike charge, doesn't accept an accessor function and only accepts a fixed setting.